

RAMANJEET SAINI

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TECHNICAL SKILLS

ML / DL: PyTorch, scikit-learn, XGBoost, NumPy, Hugging Face Transformers, QLoRA, random forests, logistic regression | **LLMs & GenAI:** LangGraph, LlamaIndex, RAG, multi-agent systems, Claude (AWS Bedrock), Llama 3, vLLM, Triton Inference Server, ONNX, FAISS | **MLOps:** AWS SageMaker, Kubeflow Pipelines, MLflow, W&B, Evidently AI, RAGAS, DeepEval, Docker, Airflow | **Data & Cloud:** Apache Spark, Kafka, DynamoDB, PostgreSQL, AWS (Lambda, EMR, Kinesis, Texttract, S3) | **Languages:** Python (expert), SQL · FastAPI, REST · GitLab CI/CD | **Domains:** ML model serving, real-time inference, recommendation systems, LLM fine-tuning, RAG pipelines, fraud detection, time-series forecasting

EDUCATION

The University of Texas at Austin

May 2023

M.S., Information Technology & Management | Concentrations: Machine Learning, Data Science | GPA: 3.96 / 4.0

Pandit Deendayal Energy University

May 2018

B.Tech., Mechanical Engineering

EXPERIENCE

Western Union

Austin, TX

Machine Learning Engineer

Aug 2023 – Present

- Architected and deployed a production **PyTorch + FAISS two-tower neural network** recommendation system driving **\$1.1M/quarter** incremental revenue; deployed via **Triton Inference Server with ONNX** at <80 ms p95 latency; evaluated with **NDCG@10, MRR, and AUC**.
- Led cross-functional deployment of a real-time fraud-detection **RAG pipeline** (LlamaIndex + Claude on AWS Bedrock), cutting analyst handle time by **26%**, raising straight-through processing by **19 pp**, and reducing false-positive rate by **14%**; monitored with RAGAS.
- Built KYC identity-verification **ML system** (97% precision / 94% recall) using **XGBoost + Claude (Bedrock) + Amazon Texttract**; saved **\$0.9M/quarter**; served via **FastAPI** with full Responsible AI compliance.
- Reduced idle float by **\$2.5M** and achieved **99% cash-availability SLA** via **time-series forecasting** with **Prophet and DeepAR on SageMaker** incorporating holiday and seasonality covariates; achieved **MAPE ≤ 7%**; mentored 2 analysts.
- Owned end-to-end MLOps for churn and CLV models (**XGBoost, random forests**) on **SageMaker + Kubeflow Pipelines**; lifted customer retention by **12 pp** and cut model release cycle by **30%** via automated retraining, MLflow + W&B, and Evidently AI drift monitoring.
- Designed agentic ML system for real-time FX pricing using **LangGraph + Claude 3**; protected a **35 bps** margin target, lifted checkout conversion by **9%**, and eliminated **80%** of manual analyst effort.

Infosys Limited

Pune, India

Data Engineer

Jul 2020 – Jun 2022

- Designed ML-powered data pipelines processing **500M+ records/week** on **Spark / AWS EMR**; integrated scikit-learn anomaly detection and Kinesis streaming; cut infrastructure costs by **34%** via cluster optimization.
- Built ML feature pipelines using **PySpark and Kafka** for personalization models (+12% throughput); optimized **AWS Lambda** inference functions, cutting serverless costs by **33%**.

Cloud Engineer

Oct 2018 – Jul 2020

- Reduced DBaaS (PostgreSQL, SQL Server, Oracle) downtime by **28%** via CloudWatch monitoring; deployed Tableau connectors via GitLab CI/CD with Python data-quality checks, improving reliability by **18%**.

PROJECTS

Snap & Track: Vision-Based Calorie Estimator (PyTorch + QLoRA) — github.com/ramanjeetsaini

2025

- Built a personal meal-logging app that identifies food items and estimates calories from a photo; trained a **PyTorch CNN** for food classification on a 100-class dataset; fine-tuned **Llama 3.1 8B with QLoRA + DPO** for portion-size reasoning, outperforming GPT-3.5 baseline by **14 F1 points** on nutritional estimation; served via **vLLM (AWQ 4-bit)** cutting GPU memory 55%; tracked experiments with W&B and evaluated with DeepEval.

Personal Music Recommender: Mood & Context-Aware Playlist Engine — github.com/ramanjeetsaini

2025

- Built a personal playlist engine that recommends songs based on current mood, time of day, and listening history; trained an end-to-end **two-tower PyTorch** retrieval model on implicit listening signals (play count, skip rate) with pairwise ranking loss; improved NDCG@10 by **18% over a matrix factorization baseline**; served top-K candidates via **FAISS** at <30 ms p95 through a FastAPI backend.